Teaching with Machines: How Instructors Characterize Generative AI's Role in Teaching Design of Interactive Systems

ANONYMOUS AUTHOR(S)

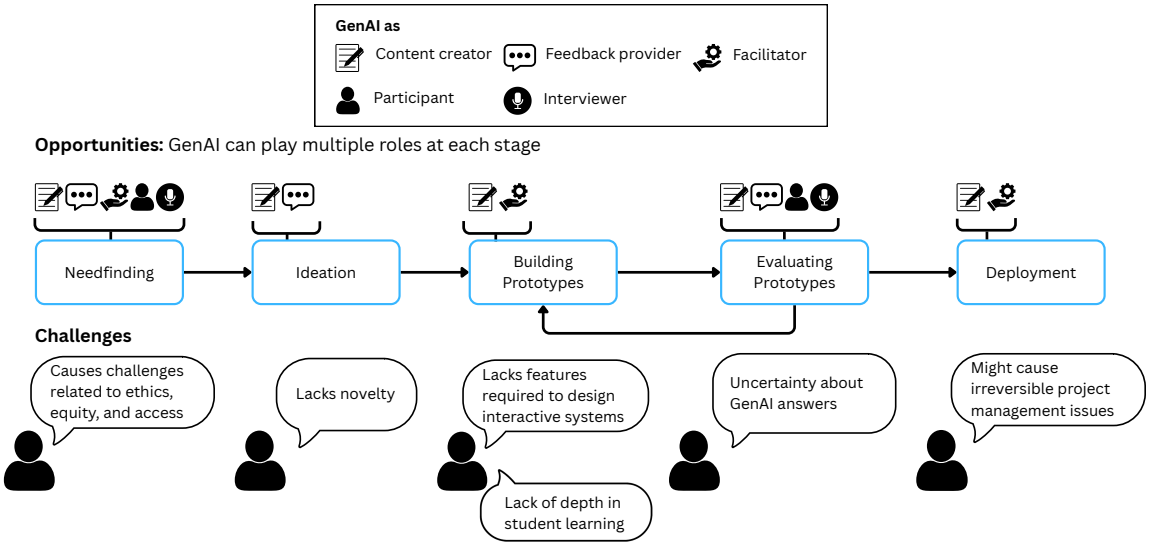

Fig. 1. Our work reports instructors' views on using GenAI when teaching how to design interactive systems. Our interview study found that GenAI can play four roles across the design process: content creator, feedback provider, facilitator, and participant/interviewer. GenAI use also brings up six challenges including concerns about learning outcomes, over-reliance, and uncertainty about GenAI outputs.

Generative Artificial Intelligence (GenAI) can potentially improve how the design process is taught in classes about designing interactive systems. How might instructors ensure that students continue to grow their design skills even when using GenAI? Our mixed methods study with seven university instructors enumerates opportunities and challenges with using GenAI when teaching how to design human-centered tools. Our findings show four roles instructors envision for GenAI: 1) content producer, 2) feedback provider, 3) facilitator, and 4) participant or interviewer. Instructors noted that while GenAI raises expectations for the speed and quality of student's work, it does not change expectations for what students should learn. Instructors also expressed concerns about GenAI's impact on student's learning, uncertainty about responses, lack of novelty, ethical and equity concerns, and current limitations of GenAI in supporting design activities. Most instructors expressed interest in redesigning their classes to include GenAI due to industry expectations and the recognition that students are likely to use GenAI.

CCS Concepts: • **Human-centered computing → Interaction design process and methods**; **Empirical studies in HCI**.

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Additional Key Words and Phrases: Design process; Generative artificial intelligence; Design education; Human-centered design

1 Introduction

Designers increasingly use Generative Artificial Intelligence (GenAI) to design interactive systems by using it throughout the design process–including needfinding, ideation, prototyping (design, implement, evaluate), and deploying the tool (Figure 1). Tools such as ChatGPT, Midjourney, DALL-E, and Copilot support brainstorming [18, 28], rapid prototyping [9, 22, 23], and feedback on prototypes [8]. As GenAI becomes embedded in design practice, designers–both novices and experts–still need to develop core design skills. These skills are necessary to critically evaluate AI-generated outputs, recognize errors, and ensure that designs remain human-centered [12, 34]. A core upcoming challenge for the technology design community is ensuring that the next generation of designers are well-trained and grounded in design concepts and methods while effectively using GenAI tools.

This challenge is particularly salient in classes that teach students how to design interactive systems. Such classes traditionally emphasize the design process, reflective thinking, and the development of design skills like user-centered design and visual design principles [13]. However, students' use of GenAI in these classes has raised concerns about superficial understanding and reduced engagement with human-centered design principles [32]. As a result, instructors face difficult questions about how to integrate GenAI into the design process while ensuring learning.

To better understand how GenAI can be integrated into classes on designing interactive systems, we conducted a mixed-methods study with instructors who teach these courses. The study focuses on instructors' perspectives, as they decide learning goals, create assignments, and shape classroom norms around GenAI use. We conducted an interview study–preceded by a survey–with seven instructors, to examine their views on opportunities and challenges of GenAI use, and their perspectives on redesigning classes to incorporate GenAI.

This paper contributes an empirical study that explores instructors' views on integrating GenAI in their classes about designing interactive systems. Instructors envision four roles for GenAI in designing interactive systems classes: content producer, feedback provider, facilitator, and participant/interviewer. GenAI can produce content–like interview guides, ideas, and prototypes–to support learning. Students can use GenAI to receive feedback on their work such as survey questions and ideas. GenAI can be a facilitator and support students in completing design tasks and learning new skills. For needfinding and prototype evaluation, students can use GenAI as a pilot participant to rehearse questions or as an assistant during interviews to track participant's responses. The results also identify six challenges instructors face when integrating GenAI, including concerns about learning outcomes, over-reliance on GenAI, and uncertainty about GenAI outputs. The study surfaced why some instructors choose to (or might decide to) redesign their classes to include GenAI and the challenges they (might) encounter in doing so.

2 Related Work

Our work builds on prior work examining technology in design education and GenAI use in designing interactive systems. Our synthesis of prior work identifies the extent of GenAI use throughout the design process and the need to understand instructor's perspective on GenAI use.

2.1 Technology in design education

Classes focusing on design of technology and interactive systems are an integral offering of many design and computer science curriculum across the world. Such classes teach concepts related to understanding user needs and context, defining requirements, ideating and prototyping solutions, evaluating the designs against requirements, and testing with

users [6, 13, 30]. New technologies influence how instructors plan and teach courses, and how students learn [2]. For example, tools such as shared digital whiteboards (like Miro), messaging platforms (like Slack), and video-conferencing systems (like Zoom) can help with essential skills in design education: brainstorming ideas, exchanging peer-feedback, documenting lecture content, and collaborating with team members [11]. Tools help with prototyping as well: students create prototypes using tools like Figma and Adobe XD, and instructors use these same tools to review and provide feedback on students' prototypes [26, 31]. Unlike many such technologies, Generative artificial intelligence's (GenAI) versatility and capabilities can influence how instructors plan and teach each stage of the design process.

GenAI is increasingly used by instructors to support their teaching practice [20]. Instructors use GenAI to support learning by providing feedback on student work and generating teaching materials such as lecture notes and quizzes [29]. They also use GenAI to make classes more engaging by customizing exercises and creating interactive activities that increase student participation [29]. Prior research at ACM DIS has identified how students use GenAI tools–like ChatGPT, Copilot, Midjourney, Adobe Firefly, and DALL-E–in design classes, highlighting the need for pen-and-paper design activities to side-step GenAI use and the importance of student reports on their use of GenAI [32]. Students' use of GenAI has been studied for the early design process, from brainstorming ideas and needfinding to designing storyboards and creating visual assets. Complementing these efforts, our research focuses on classes that teach end-to-end technology design and specifically on instructors' views of GenAI use among students learning the design process.

Integrating GenAI in teaching comes with challenges. Instructors across disciplines worry about the loss of critical thinking, academic dishonesty, and decline in social interactions among students who use GenAI [17]. To address these challenges, instructors have adopted multiple strategies. Some have increased the weight of in-person exams, while others have redesigned assignments to assess students on their problem-solving process [29]. Such efforts often require students to demonstrate their reasoning, document their steps, or explain how they used GenAI to arrive at a solution. Since instructors are already using multiple strategies to navigate the influx of GenAI in design education, understanding and characterizing these strategies can help the broader design education community anticipate challenges and opportunities of using GenAI. Understanding how instructors plan to use GenAI is critical because it shapes the design of assignments, activities, and feedback. These decisions influence whether students develop design skills effectively and avoid over-reliance on GenAI [7]. Without insight into instructors' goals and strategies, adopting GenAI could undermine learning outcomes or miss opportunities to foster creativity, collaboration, and reflective design thinking [10, 17, 27, 39]. Our work provides insights into instructors' views on using GenAI in design of interactive systems classes through a mixed methods study with instructors.

2.2 Generative artificial intelligence in designing interactive systems

Generative AI tools can support multiple stages of the design process by supporting creativity and efficiency [37]. In the needfinding stage, tools like ChatGPT can assist with qualitative analysis by performing thematic coding and helping explore initial data [40, 41]. Ideation also benefits from GenAI: research prototypes such as GANCollage can generate images aligned with a user's concepts, helping designers expand on early ideas [18, 38]. Prototyping especially benefits from integrating GenAI tools. Image-based models like DALL-E and Midjourney allow designers to quickly visualize ideas, experiment with styles, and iterate early concepts [34, 36]. Research systems such as Sketch2Prototype can convert hand-drawn prototypes into 2D images or 3D models, streamlining the transition from sketches to tangible designs [9].

Beyond generating ideas and prototypes, GenAI can also support evaluation and feedback. For instance, MemoVis, a GenAI-powered text editor, can create reference images from feedback comments, helping designers reflect on and improve their work [8]. Designers have also explored using GenAI to generate material for design fictions, providing rich speculative scenarios for testing and inspiration [3]. Practitioners rely on years of training and experience to evaluate outputs and make design decisions [34]. Despite these use cases, industry practitioners face multiple challenges when using GenAI to design interactive systems. They find it challenging to integrate GenAI into their workflows because it is often unclear what the tools can and cannot do [42]. Additionally, the outputs of GenAI vary broadly in complexity, ranging from simple suggestions to fully generated designs, which makes it difficult to anticipate how to incorporate them effectively. For example, GenAI tools can be used to generate a exploratory list of ideas and to generate a high-fidelity working prototype often with varying trade-offs in utility and novelty. GenAI can also miss human elements of design that include nuanced contextual understanding, ethical judgment, and accountability [21].

Understanding the design process is critical for evaluating and judging the correctness of GenAI outputs [21]. Junior designers who over-rely on GenAI may be unable to identify good design practices and accept GenAI outputs at face-value [34]. In educational settings, instructors must guide students in using GenAI in ways that support learning and developing design skills. Our work provides insights into the opportunities and challenges of using GenAI at each stage of the design process by learning from instructors who teach relevant classes.

3 Methods

A mixed methods study answered two research questions: **RQ1)** What are the opportunities and challenges of using GenAI in courses on designing interactive systems?; and **RQ2)** How might courses on designing interactive systems be redesigned to include the use of GenAI?

Our research team conducted a mixed-methods study with instructors of relevant design classes; the study comprised a short survey followed by a semi-structured interviews. The interviews were analyzed using reflexive thematic analysis to identify opportunities and challenges of GenAI use in designing interactive systems classes.

3.1 Participants

Participants were eligible for the study if they taught a design of interactive system class at a university. We broadly define designing interactive systems classes as classes that focus on creating and evaluating digital technology that people interact with. Such classes typically teach how to design user-centered systems by following the human-centered design process that combines principles from design and usability, psychology, and computing. Such classes take many names; some include "Interactive Device Design", "Introduction to Human-Computing Interaction", "Prototyping Studio", "Prototyping with AI", or "User-Centered Design". In such classes, students typically learn how to identify user needs, prototype solutions, test interactions, and refine designs.

Participants were recruited via the final author's existing professional network of design and human-computer interaction academics. Nineteen people were sent an email to participate, sixteen responded, and seven participated in the study (Table 1). Participants ranged in age from 25–30 years to 50 years and older. All participants teach at universities in North America. Participants had taught a design of interactive systems class multiple times (median and average = 8 times; Min = 1 time; Max = 20 times; N = 6). One participant reported extensive experience teaching related prototyping courses since 2012 but did not provide a specific number and was therefore excluded from the quantitative summary. Two participants were professors, three were associate professors, and two were assistant professors. Six participants were from public universities and one was from a private university.

ID	Position at the university	Number of time class was taught	Type of university
P1	Professor	10+	Public
P2	Associate professor	12	Private
P3	Assistant professor	3	Public
P4	Professor	Teaching since 2012	Public
P5	Associate Professor	20+	Public
P6	Assistant professor	7	Public
P7	Assistant professor	1	Public

Table 1. Overview of interview participants.

3.2 Study material

We designed our survey and interview questions based on review of existing literature and research questions. All questions were modified and iterated based on pilot study. The pilot study was conducted by the first author with the final author, who has taught a class about designing interactive systems three times, making them a suitable pilot participant. We decided not to run a pilot study with someone outside our research team partly due to the difficulty of finding time with busy academics.

3.2.1 Pre-interview survey. The pre-interview survey guided the questions asked during the interview. The survey (provided in supplementary material) covered the following topics about the instructor's teaching/class: 1) concepts and skills taught, 2) format of the class, 3) resources and tools used to create the class, 4) concerns with the use of GenAI, 5) plans of redesigning the class to include GenAI use, and 6) demographic information.

3.2.2 Interview questions. The interview questions covered four key areas which helped us answer our research questions: 1) current ways of teaching design of interactive systems, 2) ways in which instructors want to use GenAI to teach design of interactive systems, 3) different aspects of the design process that GenAI can support and cannot / should not support, and 4) ways to include GenAI in their class and challenges therein. During the interview, participants were shown the design process schematic (Figure 2). Participants were asked to reflect on their current teaching methods, opportunities of using GenAI, and challenges of using GenAI at each step of the design process.

Throughout our study, we refer to Generative AI when mentioning AI. Generative AI refers to algorithms that can create new content, such as text, images, music, or even video, based on patterns learned from existing data. For example, tools like ChatGPT can generate text, and DALL-E can create images from textual descriptions. Generative AI differs from traditional AI, which typically focuses on tasks like classification or prediction based on existing data, in

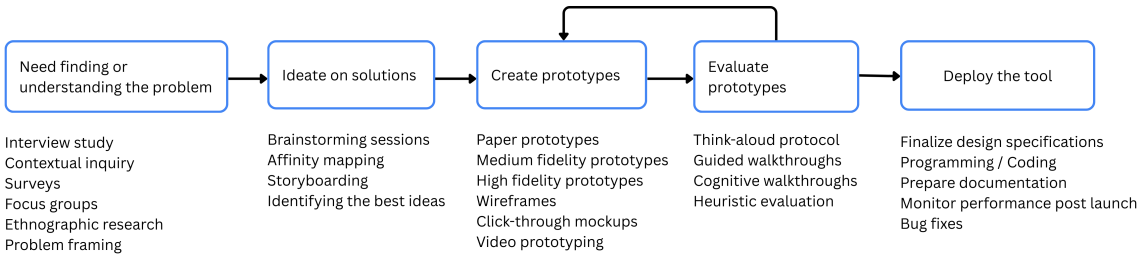

Fig. 2. Participants reflected on how they teach each stage of the human-centered design process and their views of GenAI use at each stage.

its ability to generate new outputs and artifacts. Generative AI can be used to write code, provide feedback, and to generate ideas.

3.3 Study design

Participants were contacted via email to fill the pre-interview survey and to schedule an interview time. The pre-interview survey data was analyzed to guide the interview study. The first author conducted the interviews over Zoom. All interviews were audio and video recorded with participants' consent. Participants were assured that there were no wrong answers and all responses were received without judgment. Throughout the interview, the first author asked follow up questions to gather additional details about the participant's answers. Interviews took an average of 44 minutes, median of 45 minutes (Min = 30 min, Max = 57 min). Each participant was compensated with a $30 Amazon gift card.

3.4 Data analysis

All interviews were transcribed automatically by Zoom and cleaned manually by the first author to remove filler words. The interview transcripts were analyzed by following Braun and Clarke's reflexive thematic analysis [5]. We followed the six step process of reflexive thematic analysis: 1) familiarizing ourselves with the data, 2) generating initial codes, 3) searching for themes, 4) reviewing the themes, 5) defining and naming themes, and 6) producing the report [4]. The coding of the transcript (Step 2) was done using Atlas.ti. An inductive coding approach was used to code the interview transcripts [35]. The thematic analysis (Steps 3, 4, and 5) was done using Miro–an online collaborative whiteboard platform. Consistent with reflexive approaches, the research team's perspectives and disciplinary backgrounds in human-centered design were treated as analytic resources, and reflexivity was supported through ongoing discussion throughout the analytic process. Additionally, to improve readability we have skipped repeatedly mentioning that our results draw on instructor's views and opinions throughout the results sections (*e.g.* we have minimized writing "instructors mentioned" or "instructors described").

4 Results

GenAI provides multiple opportunities and challenges when used in classes that teach designing interactive systems. We categorize the opportunities into four roles and challenges into six categories. Redesigning a class to include GenAI can trigger additional logistical concerns: it requires a lot of time and effort, and might introduce inequity among students. Overall, despite the challenges, instructors reported wanting to redesign their classes to include GenAI use.

4.1 Opportunities of using GenAI to design interactive systems

Participants reported varied opinions about GenAI use in their class in the pre-interview survey (Table 2). Three participants were open to students using GenAI at all stages of the learning process. One participant was not in favor of using GenAI. Other participants reported favoring integrating GenAI at specific stages of the design process.

Participants reported multiple opportunities of GenAI use throughout the design process. We have categorized them into four roles: content producer, feedback provider, facilitator, and participant or interviewer

4.1.1 GenAI as a content producer. Instructors see GenAI as a tool to generate materials–such as interview guides, prototypes, and documentation–that support student learning.

GenAI can be used...	Number of participants
At all stages of the learning process	3
For brainstorming or idea generation	4
To build prototypes	3
To test prototypes	2
To receive feedback	1
Only after foundational skills are learned	2
But only with full disclosure of how it assisted in the work	4
GenAI cannot be used at all	1

Table 2. Participants generally viewed GenAI as acceptable when used selectively or transparently rather than as an unrestricted tool across all stages of learning the design process. Participants could select multiple options.

Needfinding. Instructors provide interview guides in class to help students structure their interviews with people they are designed for. GenAI can help create new interview guides or improve existing ones to help students ask more relevant questions.

> *"I think they could use AI to help them write an interview guide that's within easy reach [...] Some teams take it [interview guide provided in class] over to GPT and enhance it in some way."* (P1)

> *"My students have been doing this [using GenAI] to find out what a default template is and make sure they're not missing something, and also to know when the plan that they have is trite, or cliche. So they use it [GenAI produced interview guide] to springboard off of."* (P2)

Ideation. Students can use their interactions with GenAI and its outputs to inspire ideas. When generating multiple ideas, GenAI can help identify different use cases, users, and environments. Students need to use their judgment to filter out bad ideas and select contextually appropriate approaches.

> *"One aspect of generative AI is I feel less bad now asking for more [ideas][...] What I would really like is for them [students] to think about different users, different use cases, or the same thing in a different environment."* (P2)

> *"If they [students] had already taken a stab at it [generating ideas] and they wanted to see if they forgot anything that AI could think of, maybe that would be okay."* (P7)

> *"The critical function that designers provided was in interpreting and humanizing the needs and the extension of the concepts into real solutions [...] it [GenAI] was set up to explore more broadly and go for the 30 or 40 concepts."* (P5)

Building prototypes (interactive and video). Instructors emphasize creating prototypes to make design ideas visible and interactive. GenAI can be used to quickly create high-quality digital prototypes.

> *"I saw them [students] use generative AI and it was so interesting. [...] And just within this 45 minutes to make prototypes you gotta learn how to use a Makey Makey. But this one team does Figma Make [GenAI tool] to create an interactive Dance Dance revolution. [...] their interactive screen Figma Make produced was relatively hi-fi."* (P4)

GenAI can also help edit and organize video footage recorded by the students to make a final video prototype. When a large amount of footage is collected, GenAI can help identify which videos to use.

> *"They are using generative AI to make the videos. They have a bunch of footage and then they use some kind of AI tool to make a video of it [...] They film live footage, the AI is making title slides and putting catchy music on it. I think they might even use that to help pick video? If they shoot too much video, you have to go through and pick the video."* (P2)

Evaluating prototypes. Students often create personas for target users. Such personas can later be used to evaluate the prototype. Instructors are open to students using GenAI to create personas after students create a few themselves. Students need to understand the design goals to ensure that GenAI creates appropriate personas rather than throwing up a large number of personas that provide little to no value.

> *"I want them to try themselves first, but then if they used AI to think of more personas that might be helpful, and maybe that'd be good. If the students were doing that, I would want them to have a better sense of what their design goals are because you can think of a lot of personas, and they might not actually matter for the thing you're prioritizing in your design. I would maybe want them to have clear ideas about their user population before they went and tried to generate all these [...] the main challenge is that [if] they tried to do this without really knowing why they were creating personas, you might get a lot of personas, and they might just kind of be useless."* (P7)

Programming and documentation. Some class policies allowed using GenAI for programming when programming proficiency was not evaluated as a learning outcome of the class. Students can also use GenAI to identify and fix bugs.

> *"We had a course policy where as long as it was strictly for code production, students were allowed to use AI for that. As long as they were still doing the critical decision-making themselves, as long as they were still writing any written work themselves. I think it depends on the context of the class."* (P3)

> *"You've thought about all the design decisions, and you just wanna vibe code it [...] AI would help mitigate some of those problems, because I think it's pretty good at just finding the random dumb thing that doesn't work."* (P7)

GenAI can help the documentation process by creating function headers, verifying comments and generating README files.

> *"In my opinion, [GenAI is helpful] to create function headers and even explanatory comments or even to verify that comments are up to date and it matches the code. But I haven't done a rigorous evaluation of that. That's just my own experience. It can generate these awesome READMEs."* (P4)

4.1.2 GenAI as a feedback provider. Instructors view GenAI as a tool to give feedback on student work, such as survey questions, early ideas, or prototypes. Such feedback can highlight ways to improve students' work if they engage with and act on the feedback.

Needfinding. GenAI can provide feedback on how survey questions can improve. Instructors expect students to create the initial set of questions before asking GenAI for feedback.

> *"If they had drafted survey questions, and they put it in the AI, and were like what do you think about these survey questions? [...] I would want them to make sure they tried to draft it first."* (P7)

Ideation. GenAI can be used to critique early ideas and prompt reflection on the ideas' alignment with a problem statement.

> *"They're [students] supposed to generate a bunch of ideas, and then they're asked to use LLMs as well as web search, to research their ideas and evaluate their ideas."* (P1)

Evaluating prototypes. Students often find it difficult to put themselves in the shoes of a user and identify issues with their prototypes. Such flaws are usually identified by evaluating the prototype with first-time users. GenAI can provide coarse usability analysis if provided screenshots of the system. Instructors emphasized the need for human input when evaluating prototypes.

> *"And we are seeing new systems that essentially try to act like a human and do some coarse usability analysis [...] Another example is now we have these multimodal models. So you take a screenshot of your prototype [...] it [GenAI] can start to look at things like contrast, button sizes and so on [...] I've kind of played around with that a little bit, and it's actually done pretty well."* (P4)

> *"Maybe AI can just click through the prototype just to spot obvious issues, but in the end we still want users to use it because they're designed for humans."* (P6)

4.1.3 GenAI as a facilitator. Instructors see GenAI as a tool that supports students in completing design tasks and learning new skills by summarizing data, framing problem statements, providing guidance about unfamiliar topics, and translating documentation, while leaving the design decisions to the students.

Needfinding. GenAI can be used to summarize and analyze data generated during needfinding activities.

> *"I wouldn't be surprised if they're using AI to do a little bit of the interpretation. I don't know if they do anything more sophisticated than just asking like, summarize this for me. [...] I think that's the main way they use it and, and I wouldn't have a problem with that."* (P1)

GenAI can be used to frame a student's thoughts into a well-defined problem statement.

> *"Maybe AI can be useful in terms of turning their [students'] different kinds of messy thoughts into a proper problem framing. I don't see a big improvement before GenAI or after GenAI in terms of the quality of the problem statement. I think something that AI could probably help you is make the statement itself seem grammatically correct, or makes sense, but if you actually reason on it, whether that's a real problem for the real people, in real contacts, that's something I don't see a big improvement. I think that's still the domain expertise, right?"* (P6)

Building prototypes. Students use many kinds of hardware and software components when building prototypes. When instructors struggle to help students debug and fix issues–while working with unfamiliar components–GenAI acts as a source of information.

> *"I think a lot of times people are using different hardware and different software, new libraries [...] Before, my feeling was I really couldn't do this kind of open-ended assignment if we didn't have all these resources online on YouTube, on Stack Overflow, in different user groups, and now they have generative AI, too."* (P2)

GenAI tools–like Figma Make–which are integrated into prototyping tools–like Figma–can help students learn how to use such tools. Such GenAI tools are especially useful for students who are prototyping for the first time.

> *"For students who are learning Figma, Figma Make can really show them there are these different ways of creating a prototype. So, mostly learning from AI. If AI creates a certain style, they may be like, how do I actually create this myself? [...] So they are not starting from a blank canvas and learning that. Especially for people who don't know how to use Figma."* (P6)

Deployment. Non-english speakers–who struggle to create documentation in English–can use GenAI as a translator. Project details are preserved since GenAI is translating content rather than creating it from scratch. Instructors reported that documentation created with GenAI is easily identifiable.

> *"I do think for some of my students English is not easy for them, and so sometimes they are using the generative AI tools to help them say what they want to say [...] they're saying original things clearly [...] they have very thorough documentation in whatever language they're comfortable with. They're using the AI tools to do translation of what's otherwise a pretty strong process. And I think when people are just letting the GenAI do stuff so they don't have to do the process, the difference is really stark."* (P2)

4.1.4 GenAI as a participant or an interviewer. Instructors see GenAI as a pilot participant and as an assistant during interviews. It can act as a pilot participant to rehearse questions and it can help track participant's responses.

Participant. GenAI can be used as a pilot participant during early needfinding and evaluation activities like interviews. Interviewing GenAI was described as a way for students to rehearse questioning strategies, clarify assumptions, and gain initial familiarity with a domain before conducting interviews with real users.

> *"That's the sort of thing that AI could help with. Catching some of the early problems, so that you don't have to go over those with the users [...] I wouldn't use it to replace talking to real humans that still goes to that idea of, can you simulate some of the problems? Can you anticipate and fix some of those before you waste your time with a real human being?"* (P1)

Interviewer. GenAI can support student-led interview sessions by prompting participants to perform a task or nudging them with follow up questions and keeping track of what the participant is talking about.

> *"Kind of also with the interview end of things is potentially leveraging AI to run the session. So for the think aloud protocol, for example, you could imagine the AI reminding the person to think aloud, or posting questions during the session or verbalizing them so that it keeps reminding the person to talk and what to talk about."* (P1)

> *"The other place that it [GenAI] probably can play a role is capturing what users say [...] If your user is clicking on something and talking about something, it would be good to keep track of what it is, because they'll sometimes refer to it as "pushing this button and it didn't seem to do what I wanted it to do" and so you have a transcript over here, and you have your prototype over here. I could imagine AI doing better at linking the two later when you're trying to interpret all the data."* (P1)

GenAI can also help create tasks for user studies that test all aspects of the prototype.

> *"For user studies, you usually try to come up with some tasks that you want people to do that have enough coverage of different paths and interaction flows through your system [...] AI could help you brainstorm appropriate tasks to give people in a user study."* (P7)

4.2 Challenges of using GenAI to design interactive systems

In the pre-interview survey (with pre-populated options), participants reported anticipating various challenges if students used GenAI to design interactive systems (Table 3). In the interview, participants reported multiple potential challenges with GenAI use throughout the design process that fall under six themes: 1) lack of depth in student learning, 2) uncertainty about GenAI answers, 3) lack of novelty in generated content, 4) irreversible project management issues, 5) challenges related to ethics, equity, and access, and 6) poor control for final adjustments or specific classroom needs.

Challenges of using GenAI	Number of participants who agreed
Academic Integrity & Assessment Challenges (e.g., academic dishonesty, difficulty assessing true understanding, overreliance on AI)	7
Quality & Reliability of AI-Generated Content (e.g., inaccurate or misleading information, biased or inappropriate outputs)	6
Student Behavior (e.g., reduced social interaction)	3
Privacy or Ethical Concerns	3
Student Skills & Learning Inequities (e.g., unequal access to AI tools, differences in prompt-writing skills)	2
Instructor Workload, Expectations, & Autonomy (e.g., increased workload, reduced instructor autonomy due to expectations for AI use)	1

Table 3. Survey responses showed that perceived challenges of GenAI focused on academic integrity and assessment validity. Participants raise concerns about content reliability more often than issues related to privacy, equity, or instructor workload.

4.2.1 GenAI causes lack of depth in student learning. Student's over-reliance on GenAI can lead to four concerns about their learning.

Lack of critical evaluation. Over-reliance on GenAI tools can lead to students accepting GenAI outputs without critically evaluating–carefully analyzing and judging information based on strengths and weaknesses–them. When GenAI provides feedback or evaluation, instructors expect students to validate such responses.

> *"If students just blindly take advice about script writing for an interview, or survey questions, it's just as possible for those tools to spit out bad interview questions as for my students [...] I'd feel a little annoyed if it's very obvious that they didn't lend any critical thought to their script and just had it spit it out from Gen AI."* (P3)

Lack of exploratory learning. Quick access and outputs from GenAI tools can lead to students focusing on completing specific tasks in a haste without engaging deeply with the design process itself.

> *"If the students were really, really trying to exhaust the design space, they would do one thing [with GenAI], but right now they're trying to finish an assignment, so they do something else."* (P2)

> *"I found students using generative AI [...] They simply asked, what do people need? Which is clearly just asking AI to do my project for me."* (P5)

To help students focus on the design process rather than completing the task, some instructors structure projects around real-world user research, requiring students to develop protocols, conduct interviews, and document their data collection process.

> *"I set up the projects to focus them on getting out into the world and doing this work [...] I have them come up with a protocol, have them go through the exercise of mind mapping to extract what they know, what they think they know, what they discover that they don't know about how users use a particular product, from that construct a research plan and an interview protocol. Then go out and conduct the research, record it, and they're required to share their data gathering and demonstrate that each member of the team observed and interviewed between 3 and 5 users [...] So, it's actually difficult to use generative AI."* (P5)

Lack of technical understanding. Over-reliance on GenAI for implementation can reduce understanding of how and why a solution works. For example, when GenAI is used for coding, students might have trouble debugging the code if they do not understand it.

> *"You're potentially losing out on understanding, you're potentially losing out on ownership if you don't understand what's being created. I do think that makes debugging a lot harder."* (P4)

> *"The challenges of using AI for those prototypes I think for one they are not really learning the tool. So they're relying on AI."* (P6)

Instructors noted that students who rely on GenAI for implementation can have a shallow understanding of their product and are often not able to answer questions about their work.

> *"I even had one team where the assignment was to wireframe the digital side of an experience and they worked with a generative AI tool that basically created storyboards. The storyboards all were at the level of basically an advertisement for the product. And it looked really whizzy. But then as they saw their classmates presenting, they were so mortified by the shallowness of their own presentation."* (P5)

Lack of decision-making skills. Students might rely on GenAI to make decisions that require contextual insights. For example, students might rely on GenAI to choose which idea to build their solution around. Such GenAI decisions lack students' understanding of the target demography and might not be based on insights from needfinding.

> *"I'm teaching them human-centered design which means they need to make their own decisions, but that should be as informed as possible through their need-finding and their understanding of their target demographic [...] it [GenAI] seems to be skipping over all of that, if they're using AI to make these decisions. I'm not sure AI should be making any decisions in this process."* (P3)

4.2.2 Uncertainty about GenAI answers. GenAI answers can require validation and might be a poor match for specific design stages/course content, leading to uncertainty.

GenAI content needs to be validated. GenAI output was widely described as requiring skepticism and validation because it lacks the capability to identify risks and unknowns, demonstrates biases, and lacks nuanced context. GenAI cannot accurately produce knowledge it is not trained on (and often even on data it is trained on) [1, 25]. In the needfinding stage, it is important to identify the risks and unknowns about a problem or potential idea. GenAI might not be capable of identifying relevant risks and unknown details about a problem because it is trained on documented knowledge, limiting its ability to surface unarticulated assumptions or "unknown unknowns".

> *"I don't think AI is going to do a good job at helping the thinking part of it [identifying risks and unknowns]"* (P1)

GenAI responses might be biased towards the data it is trained on. During the needfinding stage, when asked to summarize existing research or users' responses, GenAI might produce outputs that skip important outliers and reflect the normative summary. Such GenAI responses can lead to incorrect inferences.

> *"I assume that stuff [GenAI] will get better. They'll get better and not hallucinate, or do really poor summarizations of existing research. It's not very good right now. I think what we have found it to be good at is saying what the general consensus is"* (P2)

> *"If you aren't super conscious of the potential for bias, it [GenAI] might be steering you in the wrong direction."* (P4)

GenAI's modeling of people might be inaccurate. Prototype evaluation requires contextually-appropriate feedback from people who understand the task and the purpose of the prototype. GenAI's evaluation might instead surface insights that differ from those of actual target users.

> *"I don't see a lot of use because you're directly asking people and you want to hear the feedback from real people, what they think about things. I can't see a lot of things that it [GenAI] can help you with around that."* (P3)

> *"How can you replace a human's perspective of a design and their experience? [...] You cannot replace the complexity of the human mind and human experience with an AI tool, and the best way to evaluate something is to work with your unique user [...] a mom carrying a child to laugh [in her] left hand making pancakes with her right. Alexa screeching something at her, microwave beeping. That's the ultimate human complexity and we're not near that for AI to understand it, model it, and evaluate things that way."* (P4)

GenAI content might be a poor match for specific design stages/course content. Instructors expressed concern that early-stage prototypes created by GenAI might look too polished. High-fidelity prototypes invite feedback that might not be relevant to what the designer needs during early stages of design. Students can likely control the level of detail and how polished the prototype looks through prompts.

> *"We don't really want things to look very formal at the start, because you get bogged down in the details, you convey the wrong message if you're doing usability testing to your subjects [...] If they were printing off something that looked too good too early, then I think they get poorer feedback."* (P3)

> *"GenAI is at a point where it can create visualizations that are beyond the capabilities of the student and can look quite refined. Students can fall into this trap of stopping the ideation process too early or thinking that what they've produced is more finished, more refined than it is"* (P5)

GenAI might produce content that is different from the course content or what the instructor is familiar with. In such cases, instructors might not be able to help the students identify and correct mistakes.

> *"People are doing most of their programming on Raspberry Pi. When I look at the Raspberry Pi, people have folders I have never seen. They let Copilot install huge new file trees, huge new libraries. It doesn't work, because it's meant to interface with something different than they have. I'm just like I can't help you because you are so far off the path."* (P2)

Students might mistake bad ideas created by GenAI as good ideas. Students need to learn how to evaluate ideas produced by GenAI.

> *"I don't want a student to take AI-generated bad ideas as good ideas. They are not really learning. They're learning from bad material"* (P6)

4.2.3 GenAI content lacks novelty. Instructors reported that GenAI often does not produce novel and appropriate ideas. Instructors noted that incorporating ways to produce a wider range of responses can help GenAI provide greater value.

> *"We played around a lot in our research world about how to make more heterogeneous responses [ideas], and how to sort of cover a design space when you're ideating. Those things would need to come into place to get real value out of ideating with these models"* (P1)

There is a trade-off between feeling fulfilled and quickly getting the job done. Relying too much on GenAI to create content might reduce the sense of achievement one feels compared to when they create content by themselves. The lack of fulfillment could discourage students from learning how to design interactive systems.

> *"Making something is part of being human. It's a through line across every culture in society. They got great joy in producing that. And so another interesting tension is: What does that fulfillment feel like? If you are*

relying so much on Gen AI does it rob you of a sense of ownership? Or does it not matter? Is it just like any other tool and what matters is that end product, and if you feel proud of that, it's sufficient?" (P4)

4.2.4 GenAI can cause irreversible project management issues. GenAI can do irreversible significant harm when used without supervision and validation. For example, when given access to code repositories, GenAI might delete content permanently. GenAI might introduce new bugs rather than solving existing bugs when asking it to code. Such problems can happen in an instant but take a long time to overcome and solve. Instructors reported that they did not know much control people give to GenAI and how much damage it can do.

> *"We also had two students lose their GitHub repo because they let an AI do it [...] You have to do some really layered work to get through and delete everything, and they managed to let the AI do it [...] We spent so much time helping them dig their way out of that [...] we need to figure out what went wrong, and then how to set up some guardrails for that."* (P2)

To prevent such damage, instructors plan on warning future students about the dangers of GenAI use before letting them use GenAI.

4.2.5 GenAI causes challenges related to ethics, equity, and access. GenAI outputs might encourage stereotyping when used for needfinding. Stereotyping is especially harmful when the target audience of a system is an underserved or underrepresented group.

> *"I [was warned] pretty strongly on being very careful to avoid stereotyping, so I don't know if I'd encourage the students to use generative AI for research itself [...] I feel like that could get very dicey very quickly. Especially if they're targeting underserved groups or underrepresented groups."* (P3)

Students' use of GenAI can lead to reduced in-person interactions with others including classmates, users, and instructors. The lack of interaction can be harmful when students rely on GenAI to understand other people's needs instead of talking to people during the needfinding stage. Such lack of interaction and reliance on GenAI can lead to a poor or incorrect understanding of the user's problems and context.

> *"My biggest fear would be that students would become so reliant on it that they think it would be a substitute for talking to real people about their own individual needs [...] I could see people, maybe even against their own better judgment being tempted to try and gather information that way just to avoid that social awkwardness."* (P3)

GenAI is trained on a large quantity of data, some of which might be copyrighted content. Using GenAI created content might cause harm or be unfair to the people whose copyrighted content was used.

> *"Where does this training data come from? And in many cases it's violating copyright. It's stealing work from creatives. It's uncredited. It's unlicensed. It's not providing any sort of remuneration for those people that were losing their intellectual property. I definitely have objections to that."* (P3)

The quick pace of GenAI upgrades causes uncertainty in adopting a tool for a classroom. Instructors face practical issues like the need to ensure that tools taught in their classes are available long term.

> *"We're in a state of flux that I'm also a little reluctant to go hard on any one specific tool. [...] I don't know if they'll continue to offer that for free, within a month, much less 5 years from now [...] even if I do believe in the value of some of these tools, they're just so ephemeral and so likely to change that it's hard to call*

> *your exact shot on what you think will be around four years from now, five years from now, ten years from now."* (P3)

GenAI use can cause inequity among students due to three reasons. Non-WEIRD (Western, Educated, Industrialized, Rich, and Democratic) students might be at a disadvantage since most GenAI tools are trained on English data. Differences in students' prompting skills might lead to varying output qualities and learning outcomes. Finally, students who have access to better GenAI models might perform better.

> *"A lot of English versus other language, similar cultural biases there. So if your target user group is not WEIRD I really think your mileage may vary [...] And prompting these systems to improve your output at any of these stages [...] There might also be an equity argument here about who has access to these tools [...] If you give someone access to Opus 4.5 and others not, I would suspect that 4.5 is an advantage to performance."* (P4)

4.2.6 GenAI tools provide poor control for final adjustments or specific classroom needs. Most classes on design of interactive systems that teach video prototyping expect the videos to be over two minutes long. However, current, publicly available GenAI models are not capable of producing long videos. Students can potentially stitch together multiple videos but that might yield an uneven outcome.

> *"I guess AI is getting advanced enough that they could have it generate a short video. The video prototypes I'm having them do are about, like, 1-2 minutes, which is still pretty long for an AI-generated video, from what I know."* (P3)

It is difficult to control small details in prototypes created by GenAI tools like Figma Make. Such tools usually recreate the entire prototype instead of making adjustments to certain parts of the prototype. Instructors believe such challenges will be overcome in future iterations of the tool.

> *"I think this is the inherent challenge of Figma Make at the current stage. You can't do really fine control of the generation [...] you gave a description and [it] just regenerated the whole thing [...] So I don't see that as a permanent challenge. I'm sure in the next few iterations of Figma they will make it even more integrated."* (P6)

4.3 Redesigning classes on designing interactive systems to include GenAI use

Participants had mixed policies on GenAI use in their classes. Four participants allowed students to use GenAI in their class, two sometimes allowed the use of GenAI, and one participant did not allow the use of GenAI in their class. Four participants have already redesigned their classes to include GenAI use, two do not plan on redesigning their class, and one participant is currently exploring the value and use of GenAI.

When asked which GenAI tools they use during/for their lectures, three participants reported not using any GenAI tools and others used GenAI tools like ChatGPT (two), Gemini (one), Github Copilot (one), Claude (two), DALL-E (one), and Midjourney (one). One participant reported that these were the tools they are currently exploring and have not yet integrated into their course.

4.3.1 Criteria for using GenAI in classrooms. Most instructors felt that the opportunities outweigh the drawback of using GenAI. Furthermore, most believe that it is the instructors' responsibility to figure out how to responsibly include GenAI use in their classrooms. Instructors believe that in the absence of explicit rules, students might still use GenAI and it might be hard to detect such use.

> *"I think all of the benefits to me of trying to use AI in those processes takes away the drawbacks. [...] I do think it's on me as an educator to just deal with the tensions and figure out educational pathways that still utilize AI across this entire design process. I just feel so compelled to develop the next generation of HCI students [...] It's not just the hodgepodge use. I'm scaffolding it. Because I'm now explicitly injecting it, I do feel like I'm not giving up agency here. As an educator now, I have more control."* (P4)

> *"But I don't limit the use of AI at all in my class, just because detecting the use is really, really hard."* (P6)

> *"Students are going to use generative AI whether we tell them that it's okay or not. Whether we help them with it or not. They're going to use it."* (P5)

Instructors reported being open to including GenAI use in their classes as long as students learn the required design skills. GenAI is treated as a tool that helps fill gaps in users' skills–similar to other resources like Stack Overflow.

> *"I need to learn more about what you can prototype and how effectively you can prototype using AI. I wouldn't have a problem with that, as long as we could still teach how to talk to users, how to think, how to develop an intuition."* (P1)

> *"I treat it a little bit like Stack Overflow. People get ideas from other places, and I think it's fine. [...] I think there's a question of whether we should do more to let GenAI fill in some skill and gaps for people [...] The programs do a good job of helping adapt someone else's spaghetti code to do what you want."* (P2)

Policies on GenAI use in classes differ based on the type of class and the classes' learning expectations.

> *"I teach another sibling course to this one, the Intro to UI/UX. The [DIS] course that I'm mostly answering your questions in relation to is very methodology-focused and a little more academically oriented, whereas the UI/UX course is more practical [...] I will say it would make more sense for them [students] to use a Figma AI tool for that class."* (P3)

Another reason to redesign the class is the use of GenAI in the industry. Some industries expect students to know how to use GenAI when joining the workforce. In such cases, students are expected to have an understanding of both the fundamentals of the design process and how to use GenAI tools creatively. Instructors reported that it is their duty to prepare students to match industry expectations.

> *"I have heard students tell me that when they go out into the job market, they're expected to know how to use all these [GenAI] tools, or come up with clever ways to use these tools. A lot of companies are literally hiring today's grads to help modernize them. So they don't feel like they should just learn things the old-fashioned way and have the fundamentals down cold. They really feel like they're supposed to be up on these tools."* (P2)

> *"They're going to be asked to use these tools post-graduation. If we don't teach it, we're just doing a disservice [...] they're gonna be asked by employers, or it's gonna be a differentiator, or maybe even a baseline requirement for jobs. If we're not equipping them, we're doing a disservice."* (P5)

> *"People are going to use AI in their jobs."* (P7)

Some instructors reported that they plan on including GenAI in their classes incrementally rather than completely redesigning their class. Such incremental changes might help instructors test out multiple ways to include GenAI before committing to fewer strategies.

> *"I practice continuous iteration [...] So I don't think that it's a redesign. It is the small and iterative inclusion of some of these new tools [GenAI]. I think about what might be helpful, try it, see what the results are. I'm not a big fan of the kind of massive overhaul of a course."* (P5)

4.3.2 Challenges of including GenAI in classes. Instructors face three challenges when including GenAI in their classes.

Redesigning classes take a lot of time and effort. Faculty members often do not have the time to redesign classes. In such cases, they can focus on redesigning one aspect of the class instead of the entire class.

> *"Junior faculty sometimes seek advice like how can I do it all? Teaching and Research Service. If we're teaching, you can usually only modify one key thing in your course for a term."* (P4)

Course modules might need to be redesigned multiple times since the capabilities of GenAI are changing rapidly. The constant change requires additional effort each year if instructors want to keep up with the latest GenAI tools.

> *"Just between 2024 and where we're at now, so much has changed. So unfortunately I got more work ahead of me."* (P4)

Instructors also expressed concern about the effort to redesign assignments to include GenAI use. If all students don't use GenAI, the evaluation rubrics of redesigned assignments would need to be adjusted accordingly.

> *"Redesigning the assignments [...] I'd have to retool some of the assignments, and then change how I am thinking about evaluation too [...] how do I assess this? not everyone did use AI, so then what's a fair criterion to apply for assessment?"* (P7)

Including GenAI in classes might introduce inequity among students. Students' comfort and experience with GenAI tools might vary leading to inequity in outcomes. Students who know how to prompt GenAI better might be able to produce better results and learn better leading to an advantage over others. Despite the concern of inequity, instructors believe students can learn how to effectively use GenAI from each other and through experience rather than dedicating class time to teach how to use GenAI.

> *"Students have different experience with ChatGPT, or with any LLM. Some students have learned how to really leverage that for learning. And others are really first-timers. I think that can have an impact on what they get out of it and to what extent they lean on it. [...] Let's try to learn together and teach each other how to get more out of it [...] I never taught students how to use Figma. But I told them they could use it in the class. And so, I don't think I necessarily have to demonstrate the tool."* (P1)

Experience reports about previous students' use of GenAI can help guide current students use GenAI responsibly.

> *"At the beginning of this year, we had the TA [...] present the findings from the paper [about students' GenAI use] in class [...] Some people found some things that were useful, but definitely point out all these places where the students were like: I think it is not helping me, I'm not learning from these points. And so we wanted to highlight that you want to be reflective about your use."* (P2)

Instructors do not always clearly know how to include GenAI. Instructors who teach classes on designing interactive systems might not necessarily be experts in using GenAI as well; they also might not have the time/attention to keep up with the continuous changes to GenAI tools. Universities might ask instructors to include GenAI in their courses without providing proper guidelines or support to instructors. Instructors reported that they need time to understand GenAI, to tinker with it, and to figure out the best way to incorporate it into their classes.

> *"Faculty, and I'll include myself in this, we have a fairly superficial understanding of what generative AI is capable of doing, and how to draw upon that. The cause of that is that it [GenAI] is advancing so quickly. And for most of us, it's not our primary field. So we're trying to play catch-up [...] Our institutions, the universities, are demanding that we incorporate AI but they don't even know what or why and they're certainly not giving us the tools. It was this big thing that we're gonna get premium ChatGPT accounts for all faculty so that you can teach it [...] that is like buying me a laptop and saying now that you have a laptop, you need to start teaching computer engineering. That's not what I need. I need help and understanding what and how and how to teach it [...] I don't have the time to really sit [...] should I take sabbatical and spend a year just kind of figuring out how [...] we, the professoriate, educate ourselves on what we should be teaching and why around GenAI."* (P5)

> *"I don't have plans for doing that. I think most probably the reason is I don't really know good design practices of how to use AI effectively in the design process."* (P6)

Instructors are unsure about how to make time to teach about GenAI in their classes. Some classes do not have time to cover existing design phases like needfinding or deployment. Teaching about GenAI use in the design process would require instructors to drop a few topics from their current syllabus.

> *"If I were to spend more time talking about AI, what would need to come out of the syllabus to make room for that."* (P7)

The class policy on GenAI use needs to be worded appropriately to ensure that students understand what they can and cannot use GenAI for.

> *"I think I would consider being more explicit about what kinds of AI use I would support and not support [...] The language of the AI policy and the statement is really important [...] making clear what the boundaries are and how to think about that."* (P7)

4.3.3 Expectations when using GenAI in classes. GenAI use in design of interactive systems classes increase expectations on what students create. Instructors note that expectations on quality and speed of student work increases but expectations for what students learn largely remain the same when using GenAI.

> *"I expect, over time, expectations will go up in terms of the what. But it will always be the same how to think about the process, how to develop the intuition, how to ask good questions. That has to remain the same. That always has to be a point of emphasis in the teaching."* (P1)

> *"Maybe they can make prototypes faster, they can get things to work faster in some ways [...] but still dealing with the interaction, I haven't seen change in that on that side."* (P2)

> *"I think my expectations for what they learn would be consistent, but when I look at a prototype, the quality no longer maps to how much effort they put into it"* (P7)

Instructors also observed that the quality of GenAI content used by students in their classes has improved over the years. Advances in GenAI technology and students' improved prompting skills might be the reason behind the improved quality of content.

> *"The content produced over the years has significantly improved over the years [..] For one, the models improved a lot. And for two, I think students in my class are better at prompting"* (P6)

4.3.4 Reasons for not redesigning the class to include GenAI. GenAI can be used in the design process when the designer has the ability to judge the final product. Unlike students, professional designers can use GenAI in the design process since they have the required experience to judge the final product.

> *"I could see someone like a professional designer doing some of these things because… One, they know what the process looks like, and some of the things they are doing do feel like a waste of time to them, presumably, because they've done it so much. Two, they become a lot more able to judge the final product and evaluate whether it's useful or not. But for an intro-level course, the students have no heuristics for that. They can't judge whether something's good or not yet, because they just haven't built up that expertise."* (P3)

Instructors noted that they are open to allowing GenAI use in advanced courses that teach designing interactive systems.

> *"If I were to offer an advanced [DIS] class, I think I probably would be a lot more open to using those sorts of tools, because we would have established the foundational knowledge."* (P3)

4.3.5 Motivations required to consider redesigning the class to include GenAI. Instructors mention three aspects of GenAI that might further motivate them to redesign their classes to include GenAI use.

GenAI often hallucinates content and provides fake citations. Instructors would feel more confident including GenAI in their classes if it provided more reliable information and sources.

> *"The issue of hallucination is pretty much fundamental to these sorts of models [...] AI tools that would actually cite where they're drawing their particular information from could raise the reliability a whole lot for those sorts of things"* (P3)

Trust in GenAI tools being around will improve if it lasts long-term and is adopted by industry practitioners.

> *"I think standing the test of time and seeing some of these tools be around long enough that they start becoming expectations of professionals in the industry [...] That would encourage me because that means they're actually providing something useful that people want to pay for."* (P3)

Policies in universities to ensure that students use GenAI responsibly might encourage instructors to redesign their classes.

> *"The jury is still out on the wider adoption of generative AI within the department and the program. [...] if that expectation changed we were able to safely assume most students coming into a 400-level elective [advanced, upper-division electives] have received some responsible instruction around generative AI use. They've seen examples of how it can be used to help their learning versus bypassing it. If that sort of environment were created, I think I'd also be more trusting of how I could see the students using those tools."* (P3)

5 Discussion

Our research examined instructors' perceptions on the opportunities and challenges of using GenAI in courses that teach how to design interactive systems. We found that instructors view GenAI as a highly flexible resource that can take on multiple roles across the design process: content creator, feedback provider, facilitator, and participant or interviewer. At the same time, instructors expressed concerns about GenAI's impact on student's learning, particularly the risk of shallow engagement, reduced critical evaluation and decision-making skills, and over-reliance on GenAI outputs. Other challenges included uncertainty about the reliability and bias of GenAI responses, lack of novelty in

generated content, ethical and equity concerns, and current limitations of GenAI tools in supporting design activities. Despite these tensions, most instructors expressed interest in redesigning their classes to include GenAI use. Their interest is motivated by industry expectations and the recognition that students are likely to use GenAI tools regardless of class policies. Instructors noted that while GenAI raises expectations for the speed and quality of student's work, it does not change expectations for what students should learn.

5.1 Redesigning classes that teach how to design interactive systems

Most instructors reported a preference for redesigning their classes to include GenAI while maintaining learning outcomes. In our study, four of seven instructors had already redesigned their courses to include GenAI use (Section 4.3). Instructors who redesigned emphasized both the inevitability of student's use of GenAI and a sense of responsibility to support its use intentionally (e.g., P4, P5).

5.1.1 Following the human-centered design process to redesign classes. Including GenAI in classes might be done best as an iterative process–similar to designing interactive systems!–and one that instructors described as gradual and experimental rather than pushing for a single big redesign. For example, P5 described "continuous iteration" through small, incremental inclusion of GenAI rather than a "massive overhaul". Following the human-centered design process would be a good match for finding how to use novel GenAI technology in ways that meet instructors' needs from their classrooms.

Our results suggest that instructors will likely need to understand and work around potentially narrow/problematic users of such powerful tools when redesigning classes to include GenAI. Prior research notes that people find novel ways to use GenAI when designing interactive systems [24]. Instructors noted similar phenomenon in their classrooms. Specifically, they were concerned about students using GenAI to complete specific tasks–like prompting GenAI to list people's needs–at the expense of engaging deeply with the design process (*e.g.* by conducting rigorous needfinding studies). Such concerns emerged repeatedly across interviews. Instructors described students using GenAI to "finish an assignment" rather than explore the design space (P2), or directly asking AI to "do my project for me" (P5). Others noted that GenAI-supported artifacts often appeared polished but revealed shallow understanding when students were asked to explain design decisions (P5). Prior work around students' use of GenAI in the design process shows that students prioritize speed, convenience, and reduced workload [32]. Instructors in our study observed similar patterns, noting that students often used GenAI to accelerate ideation or prototyping, sometimes at the cost of exploration or reflection (P2, P6). Similarly, some university students use GenAI to complete assignments, rather than to support learning or understanding of the underlying material [14].

5.1.2 Students need to stay engaged when designing with GenAI. Instructors want to align GenAI use in their courses in ways that partially align with how students already use these tools–often to optimize for speed, convenience, and completing assignments–while ensuring that students do not over-rely on GenAI outputs. Instructors emphasized that learning to design interactive systems requires engaging with each stage of the design process, including needfinding, ideation, prototyping, and evaluation. Some instructors explicitly rejected GenAI "making decisions" on students' behalf (P3). Others described how over-reliance on GenAI undermined students' sense of ownership and ability to explain or debug their work (P4, P5). Instructors' hopes have parallels to Donald Schön's seminal work "The Reflective Practitioner" which emphasizes that expertise comes from reflecting on decisions during a task (reflection-*in*-action) and reflecting on decisions made after a task is completed (reflection-*on*-action) [33]. Designers gain expertise through the sequence of decisions made during the design process [15]. However, GenAI tools often generate final products–ideas, prototypes,

or code–without making the intermediate reasoning and decisions visible or controllable. Classes about designing interactive systems classes can benefit from GenAI tools that make the design process visible (reflection-*on*-action) and controllable (reflection-*in*-action).

5.1.3 Need for GenAI tools that support reflection-in-action. In response to instructors' concerns about students outsourcing decision-making to GenAI (P3) and prematurely converging on polished solutions (P5), GenAI tools for education can help by making design decisions explicit and interruptible. For example, instead of generating a complete prototype in response to a prompt, tools could divide prototyping into multiple steps. First, students could be asked to make explicit choices–such as number of screens, position of elements, core features, etc. Second, students could select among alternatives and justify their choices before advancing. Finally, GenAI could generate multiple prototype variants based on different decisions and ask students to explicitly compare these variants and reflect on how specific design choices shape the final outcome. By making decisions explicit, such tools would align with instructors' emphasis on learning through reflection-in-action and iteration, while preserving benefits from GenAI's ability to rapidly produce artifacts.

5.1.4 Need for GenAI tools that support reflection-on-action. Instructors consistently emphasized that GenAI should support design critique rather than replace it (Section 4.1.2), and expressed frustration when students accepted AI outputs uncritically (Section 4.2.1). To support these goals, GenAI tools could pair generation of ideas and prototypes with structured critique and reflection as successfully demonstrated in tools like PeerStudio [19]. After generating a prototype, a GenAI tool could prompt students to engage in self-assessment using rubrics that focus on alignment with the problem statement, usability, or clarity of interaction–and require students to explicitly identify limitations. The tool could then generate revised prototypes in response to the student's prompt identifying limitations, thereby making iteration an explicit and visible part of the workflow. Similar attempts could also be supported by peer assessment [19]. Such tools would help students develop critical judgment skills and position GenAI as a collaborator that responds to critique, rather than as an authority that produces complete designs.

5.2 Useful practices for using GenAI in classes that teach how to design interactive systems

Our results suggest that GenAI–like most tools–can be a great or a terrible match for design education. We suggest five practices that might help GenAI be a great match for design education.

5.2.1 Position GenAI as a support tool, not a decision maker. GenAI tools can potentially support learning when instructors reward students critically engaging with outputs and focus on process rather than speed. Most instructors in our study felt the opportunities outweigh the challenges of using GenAI, particularly when GenAI use is scaffolded rather than left unrestricted. Our interview study identified four opportunities and six challenges of using GenAI throughout the design process. Instructors in our study described students using GenAI to generate interview guides (P1, P2), ideation alternatives (P2, P5), prototypes (P4, P6), and documentation such as READMEs (P4). Interestingly, some of the opportunities recognized in our study are similar to how practitioners already use GenAI to design interactive systems. Prior work shows that practitioners use GenAI as a content creator (to create ideas, prototypes, and documentation) and as a feedback provider [34, 36]. Instructors and practitioners also share similar concerns. Instructors raised concerns about biased summaries (P2), inaccurate personas (P7), and GenAI's inability to model real-world context and human complexity (P3, P4). In prior work, practitioners expressed uncertainty in GenAI outputs and noted that GenAI might not provide contextual understanding and ethical judgment when designing interactive systems [21, 42].

GenAI can be used to support early-stage ideation/critique with added constraints that students argue about its outputs before editing and using them. Instructors in our study described deliberately constraining how GenAI was integrated into assignments to avoid shallow engagement and over-reliance on AI-generated outputs. Across interviews, instructors positioned GenAI as a tool that can support human-centered design and repeatedly emphasized that decision-making responsibilities should rest with students (P3, P7).

5.2.2 Use GenAI to accelerate producing artifacts, but not to replace the design process. GenAI can help when used to accelerate the design process based on student's decisions for tasks like analyzing data, providing feedback, and generating alternatives. When used for such tasks, GenAI can free up class time, instructor effort, and students' cognitive resources for other *sticky* activities such as problem framing, contextual inquiry, decision making, and reflecting on critique. Instructors described such use when discussing GenAI as a facilitator–for example, summarizing interview data (P1), helping frame problem statements without improving their underlying quality (P6), or assisting with debugging unfamiliar tools so instructors could support more open-ended projects (P2).

5.2.3 Leverage GenAI for breadth while requiring critique. Prior work, like Juxtapose, demonstrates that generating multiple prototypes helps exploration of solutions and facilitates discussion and reasoning [16]. Similarly, GenAI's ability to generate a large number of ideas and prototypes can support exploration. Instructors in our study reported that a large number of ideas and prototypes can be useful when students can evaluate their utility for specific user contexts rather than using those ideas without reflecting. When such engagement is absent, instructors observed challenges such as a lack of depth in learning (P5 described "whizzy" but shallow work) and increased uncertainty about the value of GenAI outputs (P3, P6).

5.2.4 Adapt GenAI use to course goals and design stages. Different courses have different learning outcomes and expectations, and may emphasize or drop different phases of the design process. Instructors in our study articulated the difference in courses by contrasting methodology-focused courses with more practice-oriented ones (P3) or introductory courses with advanced electives (P3). As a result, GenAI use needs to be adapted to each class. For example, a class focused on prototyping might prefer to use GenAI as a content producer (P4, P6), while a class focused on needfinding might prefer to develop greater techniques and guardrails around GenAI as a participant or interviewer for rehearsal and reflection (P1, P7).

5.2.5 Use GenAI for structural checks of work, not as a substitute for user engagement. GenAI can help improve the structure of ideas, surveys, interviews, and prototypes when access to actual users is limited. Instructors described GenAI acting as a pilot participant to help students rehearse interview questions, catch early issues, or explore a domain before engaging real users (P1). Instructors were explicit in noting that GenAI cannot replace human perspectives or lived experience, particularly in complex contexts (P3, P4).

5.3 Limitations and future work

Our work uncovers instructors' perspectives on the use of GenAI in design education rather than evaluating the technical capabilities of GenAI systems themselves. As a result, our findings reflect how instructors perceive and experience GenAI. Future work could directly compare instructors' expectations and concerns with the actual capabilities and limitations of GenAI tools used in design practice. All participants in our study were based in the United States. Teaching methods, institutional constraints, and expectations around GenAI can vary substantially across cultural and educational contexts. Such differences will shape both opportunities and challenges of GenAI use. Future studies need to include

instructors with global representation to examine how these factors influence pedagogical integration in their context. Finally, our study included a small-but-fair number of participants due to instructors' time constraints. Around the fifth interview in our study, we observed repetitive findings and diminishing returns in new insights across interviews, suggesting thematic saturation. Future work could validate and extend our findings with larger or longitudinal samples.

6 Conclusion

The paper describes four roles that instructors' envision for GenAI when designing interactive systems: content creator, feedback provider, facilitator, and participant or interviewer. Instructors also expressed six concerns due to GenAI use throughout the design process. Despite the challenges, most instructors expressed an interest in redesigning their classes to include GenAI.

References

[1] Maryam Alavi, Dorothy Leidner, and Reza Mousavi. 2024. Knowledge management perspective of generative artificial intelligence (GenAI). *Alavi, Maryam* (2024), 1–12.

[2] Tony Bates and Gary Poole. 2003. Effective teaching with technology in higher education: Foundations for success. (2003).

[3] Mark Blythe. 2023. Artificial design fiction: Using ai as a material for pastiche scenarios. In *Proceedings of the 26th International Academic Mindtrek Conference*. 195–206.

[4] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.

[5] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative research in sport, exercise and health* 11, 4 (2019), 589–597.

[6] Bill Buxton. 2010. *Sketching user experiences: getting the design right and the right design*. Morgan kaufmann.

[7] Julio Cabero-Almenara, Antonio Palacios-Rodríguez, María Isabel Loaiza-Aguirre, and Paola Salomé Andrade-Abarca. 2024. The impact of pedagogical beliefs on the adoption of generative AI in higher education: predictive model from UTAUT2. *Frontiers in Artificial Intelligence* 7 (2024), 1497705.

[8] Chen Chen, Cuong Nguyen, Thibault Groueix, Vladimir G Kim, and Nadir Weibel. 2024. Memovis: A GenAI-powered tool for creating companion reference images for 3d design feedback. *ACM Transactions on Computer-Human Interaction* 31, 5 (2024), 1–41.

[9] Kristen M Edwards, Brandon Man, and Faez Ahmed. 2024. Sketch2Prototype: rapid conceptual design exploration and prototyping with generative AI. *Proceedings of the Design Society* 4 (2024), 1989–1998.

[10] Ali Fulsher, Marianna Pagkratidou, and Panayiota Kendeou. 2025. GenAI and misinformation in education: a systematic scoping review of opportunities and challenges. *AI & SOCIETY* (2025), 1–13.

[11] Mafalda Gamboa and Sara Ljungblad. 2022. Designerly Ways of Knowing in HCI Education: A Case Study of a Peer Community-Based Studio. *Frontiers in Computer Science* 4 (2022), 793968.

[12] Camille Grange, Theophile Demazure, Mickael Ringeval, Simon Bourdeau, and Cedric Martineau. 2026. The Human-GenAI Value Loop in Human-Centered Innovation: Beyond the Magical Narrative. *Information Systems Journal* 36, 1 (2026), 29–51.

[13] Saul Greenberg. 1996. Teaching human computer interaction to programmers. *Interactions* 3, 4 (1996), 62–76.

[14] Jan Henrik Gruenhagen, Peter M Sinclair, Julie-Anne Carroll, Philip RA Baker, Ann Wilson, and Daniel Demant. 2024. The rapid rise of generative AI and its implications for academic integrity: Students' perceptions and use of chatbots for assistance with assessments. *Computers and Education: Artificial Intelligence* 7 (2024), 100273.

[15] Björn Hartmann, Scott Doorley, and Scott R Klemmer. 2008. Hacking, mashing, gluing: Understanding opportunistic design. *IEEE Pervasive Computing* 7, 3 (2008), 46–54.

[16] Björn Hartmann, Loren Yu, Abel Allison, Yeonsoo Yang, and Scott R Klemmer. 2008. Design as exploration: creating interface alternatives through parallel authoring and runtime tuning. In *Proceedings of the 21st annual ACM symposium on User interface software and technology*. 91–100.

[17] Emma Harvey, Allison Koenecke, and Rene F Kizilcec. 2025. " Don't Forget the Teachers": Towards an Educator-Centered Understanding of Harms from Large Language Models in Education. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–19.

[18] Jingoog Kim and Mary Lou Maher. 2023. The effect of AI-based inspiration on human design ideation. *International Journal of Design Creativity and Innovation* 11, 2 (2023), 81–98.

[19] Chinmay Kulkarni, Yasmine Kotturi, Michael S Bernstein, and Scott Klemmer. 2016. Designing scalable and sustainable peer interactions online. In *Design Thinking Research: Taking Breakthrough Innovation Home*. Springer, 237–273.

[20] Sam Lau and Philip Guo. 2023. From" Ban it till we understand it" to" Resistance is futile": How university programming instructors plan to adapt as more students use AI code generation and explanation tools such as ChatGPT and GitHub Copilot. In *Proceedings of the 2023 ACM Conference on International Computing Education Research-Volume 1*. 106–121.

[21] Jie Li, Hancheng Cao, Laura Lin, Youyang Hou, Ruihao Zhu, and Abdallah El Ali. 2024. User experience design professionals' perceptions of generative artificial intelligence. In *Proceedings of the 2024 CHI conference on human factors in computing systems*. 1–18.

[22] Zhaohui Liang, Xiaoyu Zhang, Kevin Ma, Zhao Liu, Xipei Ren, Kosa Goucher-Lambert, and Can Liu. 2025. StoryDiffusion: How to support UX storyboarding with Generative-AI. In *Proceedings of the 27th International Conference on Multimodal Interaction*. 135–144.

[23] Vivian Liu, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2023. 3DALL-E: Integrating text-to-image AI in 3D design workflows. In *Proceedings of the 2023 ACM designing interactive systems conference*. 1955–1977.

[24] Monica Maceli, Nancy Smith, and Gatha Bhakta. 2024. Incorporating unanticipated uses of generative AI into HCI education. In *Proceedings of the 6th Annual Symposium on HCI Education*. 1–7.

[25] Negar Maleki, Balaji Padmanabhan, and Kaushik Dutta. 2024. AI hallucinations: a misnomer worth clarifying. In *2024 IEEE conference on artificial intelligence (CAI)*. IEEE, 133–138.

[26] Jonas Oppenlaender, Elina Kuosmanen, Andrés Lucero, and Simo Hosio. 2021. Hardhats and bungaloos: Comparing crowdsourced design feedback with peer design feedback in the classroom. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–14.

[27] Hyanghee Park and Daehwan Ahn. 2024. The promise and peril of ChatGPT in higher education: opportunities, challenges, and design implications. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–21.

[28] Xiaohan Peng, Janin Koch, and Wendy E Mackay. 2024. Designprompt: Using multimodal interaction for design exploration with generative ai. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 804–818.

[29] James Prather, Juho Leinonen, Natalie Kiesler, Jamie Gorson Benario, Sam Lau, Stephen MacNeil, Narges Norouzi, Simone Opel, Vee Pettit, Leo Porter, et al. 2025. Beyond the hype: A comprehensive review of current trends in generative AI research, teaching practices, and tools. *2024 Working Group Reports on Innovation and Technology in Computer Science Education* (2025), 300–338.

[30] Jenny Preece, Yvonne Rogers, Helen Sharp, David Benyon, Simon Holland, and Tom Carey. 1994. *Human-computer interaction*. Addison-Wesley Longman Ltd.

[31] Emma J Rose, Craig M Macdonald, and Cynthia Putnam. 2022. Teaching Design Systems: Towards a flexible and scalable model for the UX classroom. In *Proceedings of the 40th ACM International Conference on Design of Communication*. 107–113.

[32] Hauke Sandhaus, Qiuquan Gu, Maria Teresa Parreira, and Wendy Ju. 2025. Co-Designing with Transformers: Unpacking the Complex Role of GenAI in Interactive System Design Education. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. 1228–1243.

[33] Donald A Schön. 2017. *The reflective practitioner: How professionals think in action*. Routledge.

[34] Macy Takaffoli, Sijia Li, and Ville Mäkelä. 2024. Generative AI in user experience design and research: how do UX practitioners, teams, and companies use GenAI in industry?. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 1579–1593.

[35] David R Thomas. 2003. A general inductive approach for qualitative data analysis. (2003).

[36] Severi Uusitalo, Antti Salovaara, Tero Jokela, and Marja Salmimaa. 2024. " Clay to Play With": Generative AI Tools in UX and Industrial Design Practice. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 1566–1578.

[37] Priyan Vaithilingam, Ian Arawjo, and Elena L Glassman. 2024. Imagining a future of designing with ai: Dynamic grounding, constructive negotiation, and sustainable motivation. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 289–300.

[38] Qian Wan and Zhicong Lu. 2023. Gancollage: A gan-driven digital mood board to facilitate ideation in creativity support. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 136–146.

[39] Peter Wulff and Marcus Kubsch. 2025. Learning against the machine: the double edged sword of (Gen) AI in STEM education. *International Journal of STEM Education* 12, 1 (2025), 66.

[40] Ziang Xiao, Xingdi Yuan, Q Vera Liao, Rania Abdelghani, and Pierre-Yves Oudeyer. 2023. Supporting qualitative analysis with large language models: Combining codebook with GPT-3 for deductive coding. In *Companion proceedings of the 28th international conference on intelligent user interfaces*. 75–78.

[41] Lixiang Yan, Vanessa Echeverria, Gloria Milena Fernandez-Nieto, Yueqiao Jin, Zachari Swiecki, Linxuan Zhao, Dragan Gašević, and Roberto Martinez-Maldonado. 2024. Human-ai collaboration in thematic analysis using chatgpt: A user study and design recommendations. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.

[42] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining whether, why, and how human-AI interaction is uniquely difficult to design. In *Proceedings of the 2020 chi conference on human factors in computing systems*. 1–13.

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